

## Options

Set defaults with `#calepin.setup` and override per call with `#calepin.chunk(...)` or `#calepin.inline(...)`.

### Global + chunk options

These options can be configured in `#calepin.setup` as document-wide defaults. Use one setup block to define defaults for all chunks unless overridden per chunk.

| Option            | Default  | Meaning                                                                                                                                                                                                                   |
|-------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| echo              | true     | Show the chunk's source code in the rendered document.                                                                                                                                                                    |
| eval              | true     | Execute the code. When false, nothing runs and no output is produced (the source can still be shown via echo).                                                                                                            |
| error             | false    | When true, capture an execution error and render it as output. When false, an error in the chunk aborts the build.                                                                                                        |
| warning           | true     | Include warnings emitted by the engine in the output. When false, they are suppressed.                                                                                                                                    |
| message           | true     | Include informational messages emitted by the engine (for example R's <code>message()</code> output). When false, they are suppressed.                                                                                    |
| results           | "render" | How results are shown: render (pretty display of values, images, and tables), verbatim (raw output in a code block), typst (treat output text as Typst markup and render it), or hide (run the code but omit its output). |
| fig-device-format | "svg"    | Format for figure files written by the engine: svg, png, jpeg (alias jpg), or pdf. Diagram engines always emit svg regardless of this setting.                                                                            |
| fig-device-dpi    | 150      | Resolution in dots per inch for raster formats (png, jpeg). Ignored for vector formats (svg, pdf).                                                                                                                        |
| fig-device-width  | 6        | Width of the plotting device, in inches.                                                                                                                                                                                  |
| fig-device-height | "auto"   | Height of the plotting device, in inches. auto derives it from the width and fig-device-aspect.                                                                                                                           |
| fig-device-aspect | 0.618    | Height-to-width ratio used when fig-device-height is auto: device height = fig-device-width × fig-device-aspect.                                                                                                          |
| fig-width         | "70%"    | Width of the figure as rendered in the document. Accepts a Typst length or ratio (for example 70% or 12cm) or auto.                                                                                                       |
| fig-height        | "auto"   | Height of the figure as rendered in the document. Accepts a Typst length or auto.                                                                                                                                         |

|                |          |                                                                                                                         |
|----------------|----------|-------------------------------------------------------------------------------------------------------------------------|
| fig-align      | "center" | Horizontal alignment of the figure in the document: left, center, or right.                                             |
| fig-responsive | true     | HTML output only: allow the figure to shrink to fit narrow viewports (sets max-width: 100%). No effect on paged output. |

## Chunk-only options

These options are only understood on individual chunks; they are not valid as keys in `#calepin.setup`. The figure caption and layout options live here because they apply to one chunk's specific output.

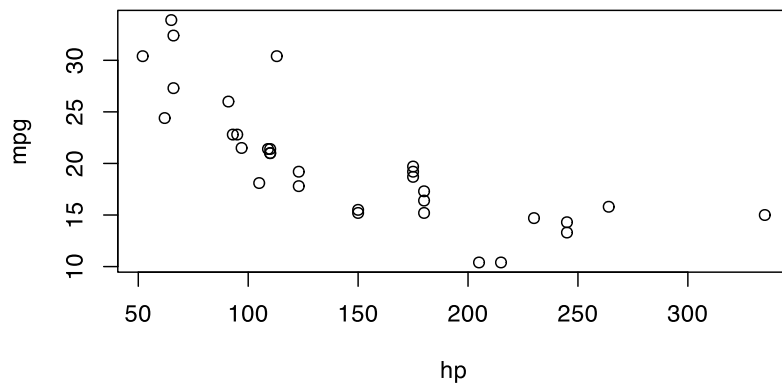
| Option             | Default    | Meaning                                                                                                                                                           |
|--------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| engine             | inferred   | Force the engine/language for this chunk instead of inferring it from the fence or surrounding context.                                                           |
| body               | from fence | Provide the raw code body directly instead of writing a fenced block.                                                                                             |
| label              | auto       | Assign a stable chunk identifier used for cross-references and result lookup.                                                                                     |
| fig-caption        | none       | Caption text for the figure. When set, the output is wrapped in a numbered figure that can be cross-referenced.                                                   |
| fig-cap-location   | "auto"     | Where the caption sits relative to the figure: top, bottom, or margin. auto uses Typst's default placement.                                                       |
| fig-alt-text       | none       | Accessibility (alt) text for generated images. Empty when unset.                                                                                                  |
| fig-subcaptions    | none       | Per-panel captions for a multi-image chunk, given as an array of strings (one per image, in order).                                                               |
| fig-layout-columns | "auto"     | Column layout for a multi-image chunk: an integer number of equal columns, an array of explicit track sizes, or auto to choose a count from the number of images. |
| fig-layout-rows    | "auto"     | Row layout for a multi-image chunk: an integer number of equal rows, an array of explicit track sizes, or auto.                                                   |

## Quarto-style header options

Quarto-style chunk headers are supported by placing option lines that begin with `#|` at the top of a code fence. These lines are parsed as the same options as `#calepin.chunk` options. For example,

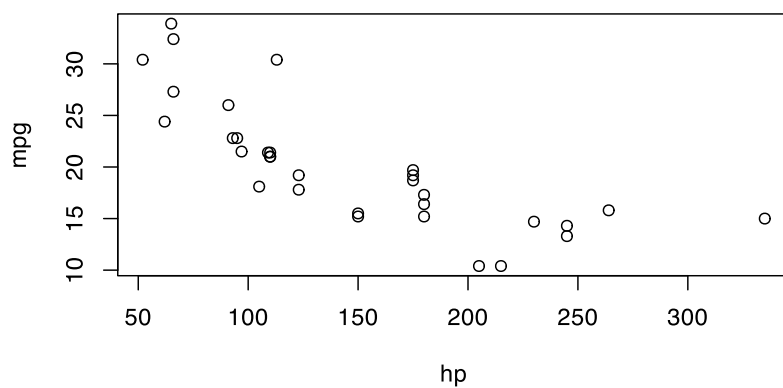
```
```r
#| fig-align: left
plot(mpg ~ hp, data = mtcars)
```
```

```
plot(mpg ~ hp, data = mtcars)
```



```
```\r\n#| fig-align: right\r\nplot(mpg ~ hp, data = mtcars)\r\n```\r\n
```

```
plot(mpg ~ hp, data = mtcars)
```



The main disadvantage of Quarto-style headers is how they behave outside *Calepin*. If you compile the .typ file directly with typst, the #| lines are displayed as configuration artifacts inside the code block. When options are specified as arguments to `#calepin.chunk(...)`, direct typst compilation shows the nice unevaluated code chunk instead.

Note that chunk options have different names in *Calepin* and *Quarto*. Some aliases are supported, but using them is not recommended and *Calepin* will emit a warning when it encounters an unsupported option name.

- out-width maps to fig-width
- out-height maps to fig-height
- out-align maps to fig-align
- fig-alt maps to fig-alt-text
- fig-subcap maps to fig-subcaptions

- fig-format maps to fig-device-format
- fig-dpi maps to fig-device-dpi
- layout-ncol maps to fig-layout-columns
- layout-nrow maps to fig-layout-rows